

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1500

Grantsburg Diesel 138 kV
June 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1500. The Interconnection Customer proposes a 58.8 MW battery energy storage facility interconnecting at the Grantsburg Diesel 138 kV substation in the West Virginia area. The request was received on June 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 391 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$916,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WESTONE SW YD-POPLAR BLUFF GENERATING STATION 138kV	HUBBADSTOWN-TRISTATE 345kV	124.2	280	5.6
POPLAR BLUFF GENERATING STATION 345 kV/138 kV Auto	HUBBARDSTOWN-TRISTATE 345kV	118.2	418	10.4
SHERIDAN-BLAND 138kV	RED MAPLE-VLY VIEW 138kV L	100.9	209	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WESTONE SW YD-POPLAR BLUFF GENERATING STATION 138kV	Rebuild line with higher-capacity conductor	\$16,288,000	2% (\$404,000)
POPLAR BLUFF GENERATING STATION 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$15,287,000	3% (\$436,000)
BLAND-SHERIDAN 138kV	Reconductor line	\$2,012,000	4% (\$76,000)
	Total allocated to project		\$916,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.